

		$\rightarrow u_X + u_Y = v_X + v_Y$
6	C	Raw Power input = Rate of GPE converted to Electrical Energy $= \frac{mgh}{t} = \frac{\rho Vgh}{t}$ $= \frac{1000(6.0)(9.81)(80)}{1}$ $= 4.78088 \text{ MW}$ Since Efficiency = $\frac{P_{out}}{P_{in}} = 0.60$ Then $P_{out} = 0.60(4.78088) = \mathbf{2.8 \text{ MW}}$

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Qn	Ans	Suggested solution
7	D	$(2 - \sqrt{2})^2 = 4 - 2\sqrt{2}$